

High Performance Active Battery Management System with Multi-Winding Transformer

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Abstract—This paper proposes an active balancing Battery Management System (BMS) that allows achieving high equalization efficiency through a Multiple Active Bridge (MAB), which interfaces cells or modules of a battery pack to a multi-winding transformer. The proposed solution dynamically transfers charge from the cells with high voltage to the ones with lower voltage with the aim of achieving a charge equalization. Numerical and experimental results confirm the effectiveness and the potentiality of the proposed solution that allows performing a simultaneous energy transfer among the cells of a battery pack.

Index Terms—Battery pack, Active balancing, BMS, EV, Multi-winding transformer, Multiple active bridge, Charging

I. INTRODUCTION

Nowadays, the increasing interest in lithium-ion battery systems (LIBs) has led to remarkable developments in Battery Management System (BMS) technology. Electrified vehicles and smart grids are two of the applications that are benefiting from the constant improvement in LIBs in terms of energy/power density and control [1]. These technological developments are progressively pushing the LIBs application in electric and hybrid vehicles with improved driving range, as well as grid-tied energy storage systems for integration of renewable energy, peak shaving and load leveling [2]–[5]. The enhancements in the BMS field are targeting cost-effective solutions capable of increasing the pack efficiency, performance, and lifetime while enforcing the operation within the safe voltage, current, and temperature ranges.

A battery pack is usually composed of cells in series and parallel to achieve the voltage and capacity required by the application. Battery cells in series may gain different voltages due to manufacturing inaccuracy, thermal gradients, and aging [6]. This voltage unbalance can: cause over-charge or over-discharge of some of the cells; limit the usable capacity of the whole pack, since the charge or discharge process will need to be interrupted if one of the cells reaches the maximum or minimum voltage limit. To increase the usable cell capacity and reduce the hazard and degradation of the cells due to over-charge/discharge, an equalization circuit is included in the BMS to balance the cell voltages when required. Several BMS architectures have been proposed in literature showing different trade-off among efficiency, equalization speed, hardware

and software complexity, size, cost, voltage and current stress [7]–[27]. The solutions based on passive balancing technique dissipate the extra charge of the cells on a power resistor placed in parallel with each cell. The resistors are activated by a transistor/mosfet controlled by a micro-controller. These architectures are widely used in automotive and industrial application and are demonstrated to be a cost-effective solution to reduce over-charging. However, their disadvantages include low equalization efficiency, unidirectionality of the power flow, and thermal issues when high equalization speed is required or large cells are considered due to the Joule's effect on the balancing resistors [28]–[30]. Higher performance can be achieved with non dissipative balancing techniques those allow for transferring energy from a cell to another cell, string or whole pack. This energy transfer can be unidirectional or bidirectional, thus a multitude of different architectures and control methods have been proposed in literature. The cell-to-cell active balancing technique is considered by many as a promising technology since it has the potential of increasing the equalization efficiency and speed limiting the number of components intercepted during the balancing action [14], when compared to cell-to-string and cell-to-pack strategies. There are two main categories: adjacent cell-to-cell equalization methods and direct cell-to-cell equalization methods. The former methods are usually based on bi-directional dc-dc converters (such as buck-boost), the cell with higher voltage is discharged on the adjacent ones until the equalization is reached. For large battery packs, these methods result very slow and expensive.

The direct cell-to-cell equalization methods are based on sharing the energy between the cells through a common storage element, such as a flying capacitor, an inductor, or a multi-winding transformer. When comparing these methods, the multi-winding approach results more convenient since it potentially allows for achieving higher equalization speed, and good control of the equalization dynamic, at the expenses of increasing complexity and cost of hardware and software [14], [31]. The equalization efficiency of the multi-winding solution can be limited by the conduction losses of the rectification stages, if not properly addressed by the control strategy [15].

In this paper, a new architecture and control of an active

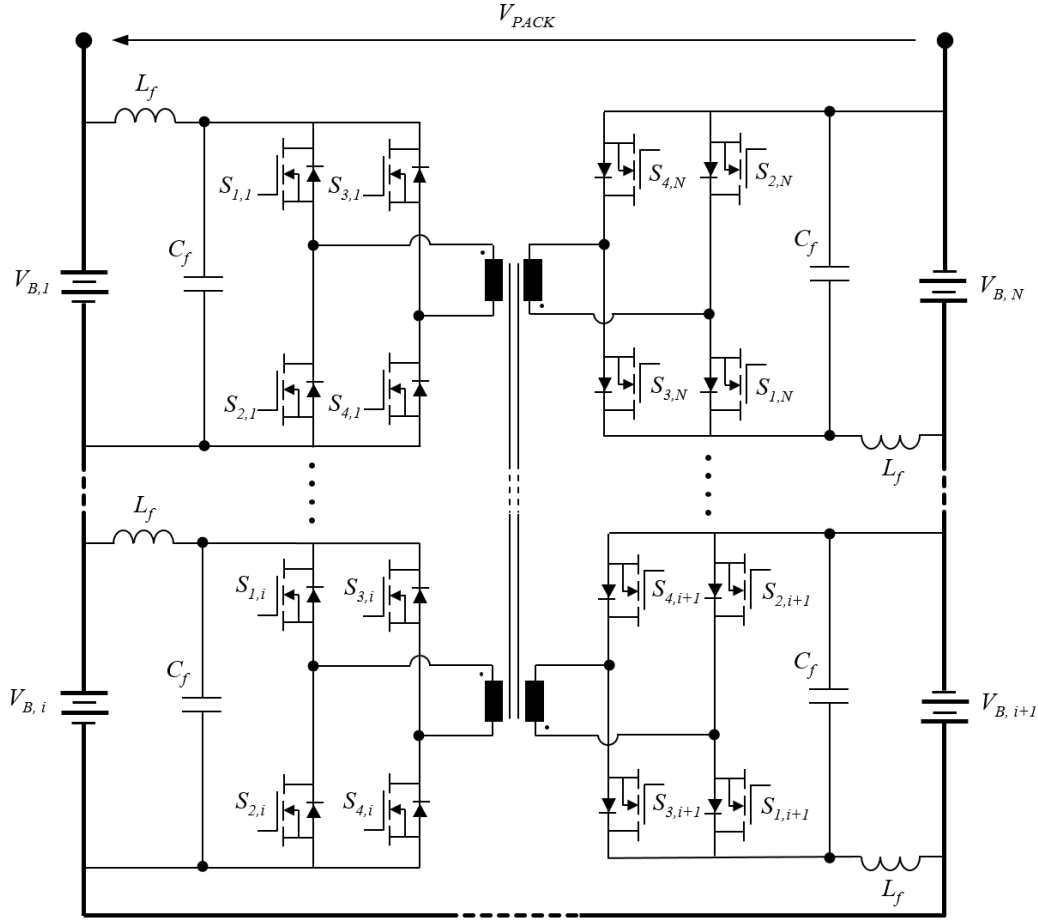


Fig. 1. Structure of the proposed active equalization circuit.

equalization circuit based on multi-winding transformer is proposed. Every cell is equipped with a H-bridge and is then connected to a multi-winding transformer, forming a Multi Active Bridge (MAB). This circuit topology has been previously proposed for medium-voltage grid-connected system [32], while it is here exploited for BMS active balancing circuit since it allows for the modular and flexible cells-to-cells operation, with the potential of increasing the equalization speed and efficiency. The complexity of the hardware architecture is here well balanced by a simple and reliable control strategy. By assuming a fixed phase shift equal to π , every H-bridge can be controlled by a common PWM signal that allows for the inversion or rectification of the cell energy depending on the voltage difference with respect to the other cells. A numerical simulation and a prototype of the equalization circuit have been implemented to demonstrate the feasibility and performance of the proposed solution.

II. STRUCTURE AND CONTROL OF THE ACTIVE EQUALIZATION CIRCUIT

The structure of the proposed active balancing circuit is based on a MAB [32], in which every cell is equipped with

a H-bridge then connected to a medium-frequency multi-winding transformer, as reported in fig. 1. Considering a battery pack composed of N cells connected in series, the proposed balancing circuit requires N inductor filters (L_f), N H-bridges with snubber capacitors (C_f) and a transformer with N windings which have the same number of turns. Despite the hardware complexity due to the presence of a multi-winding transformer and the H-bridges, the proposed structure has several advantages, such as short and bi-directional energy transfer path between cells, modularity, scalability, and the capability of limiting the equalization currents by means of the transformer leakage inductance. High equalization speed can be achieved by simultaneously transferring energy among several cells of the battery pack. Indeed, a cells-to-cells balancing can be performed. The issue of managing a large number of switches is solved thanks to a proper control algorithm based on a phase-shift modulation technique using a common PWM signal (S). A fixed phase shift (equal to π) is implemented between two legs of each H-bridge, allowing for the generation of a single PWM signal for controlling the N power conversion units. The PWM signal is characterized by a duty cycle equal to 50% and a constant

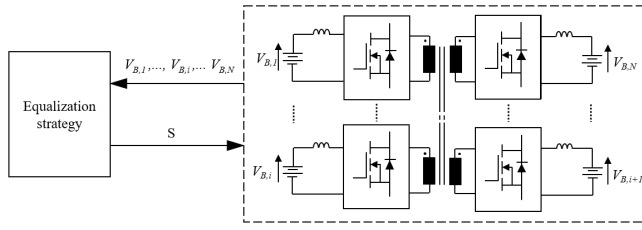


Fig. 2. Scheme of the control strategy.

frequency.

The control scheme of the proposed architecture is shown in fig. 2. The control section acquires the cell voltages $\{V_{B,1}, \dots, V_{B,i}, \dots, V_{B,i+1}, \dots, V_{B,N}\}$ of the battery string and verifies if the cells are within the desired voltage band Δ_v (e.g. voltage difference between the highest and lowest cell voltages). If the cell voltages are unbalances, the equalization strategy enables the PWM signal S to be provide to all the drivers of the N H-bridges. Thus, depending on the cell voltage levels, a current flow will be established from the cells with higher voltage (conversion stages working as an inverter) to the cells with lower voltages (conversion stages working as a rectifier). In principle, the control of the rectification stage could be disabled taking advantages of the free-wheeling diodes. However, to reduce the losses due to the diode forward voltages and to minimize the amplitude of Δ_v , the rectification stage has been controlled as a synchronous rectifier through the same PWM signal S . The balancing action is stopped when the cells voltages are all contained in the desired band.

It is important to mention that this balancing architecture can operate in both charging and discharging, and that it does not require a very accurate estimation of the state of charge SoC , since the equalization of the cells is automatically and dynamically guaranteed by the energy transfer across the transformer and by the unique control signal S . However, the selection and design of the electronic components and transformer have a great impact on the accuracy of the voltage balancing as well as on the equalization efficiency.

III. NUMERICAL RESULTS

The MAB equalization circuit and related control strategy have been implemented in MATLAB[®] using Simulink-Simscape. The battery pack used for the numerical analysis is composed by a string of four LG 18650HG2 cylindrical cells. The simulation environment includes accurate models of the battery cells, multi-winding transformer, and H-bridges, including detailed models of the power losses of the components. A first-order equivalent circuit model of the battery cell has been calibrated by means of experimental data at different charging and discharging C-rates; while, datasheet parameters have been considered for transformer and the conduction and switching losses of power switches. Table I reports the parameters of the numerical simulator.

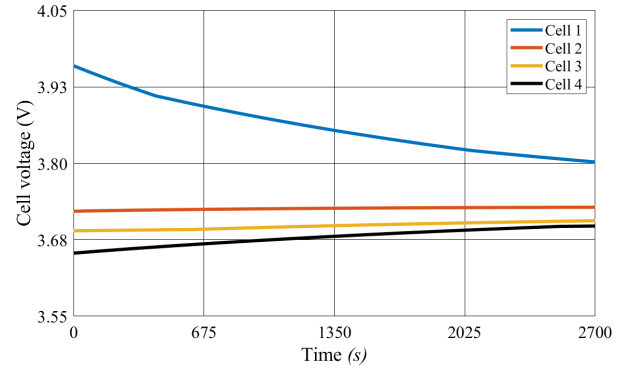


Fig. 3. Numerical results: cell voltages during the equalization process.

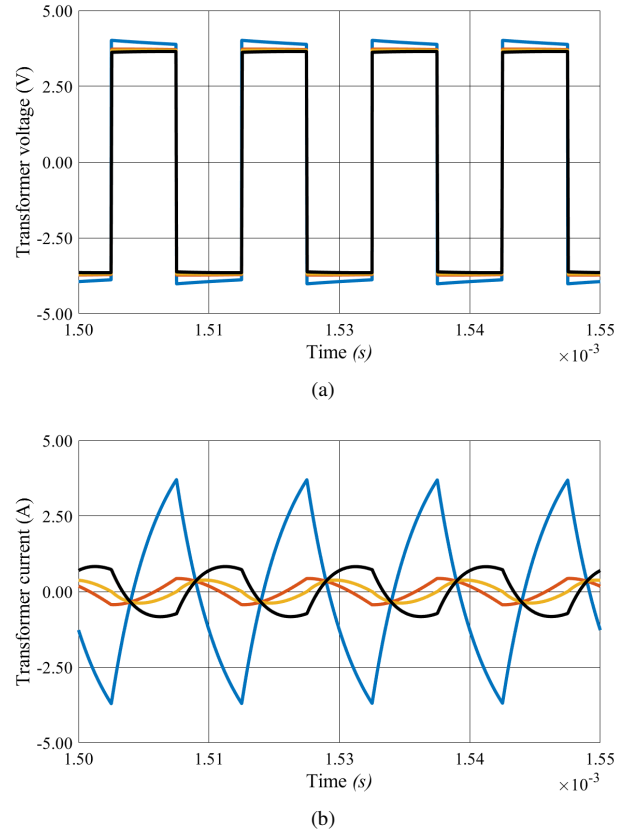


Fig. 4. Numerical results: (a) Voltages and (b) currents for each winding of the transformer. Blue: winding 1, Red: winding 2, Yellow: winding 3, Black: winding 4.

The performance of the proposed solution has been evaluated in the idle state of the battery pack with a Δ_v equal to 100 mV. Figure 3 reports the cell voltages during the equalization process of the pack in which cell 1, being the one having higher voltage, is discharged on the other cells until reaching the equalization balance. The initial state of charge of the cells is $\{SoC_1=87.6\%, SoC_2=46\%, SoC_3=44\%, SoC_4=30\%\}$. The waveforms of the currents and voltages of the multi-winding transformer during the modulation period (equal to

TABLE I
PARAMETERS OF THE PROPOSED MAB EQUALIZATION CIRCUIT.

Parameter	Value
<i>Battery cell LG 18650HG2</i>	
Typical Capacity	3 Ah
Energy Density	235 Wh/kg
Nominal Voltage	3.6 V
Cut-Off Voltage	2 V
Nominal Charging Current	1.5 A (@0.5C)
Maximum Charging Current	4 A (@1.3C)
Maximum Discharging Current	20 A (@6.6C)
<i>Infineon MOSFET IRLB8721PBF1</i>	
Drain-to-source on resistance	8.7 mΩ
Switching frequency	80 kHz
Rise time	93 ns
Fall time	17 ns
<i>Multi-winding transformer</i>	
Inductance base	3.4 μH
Rated current base	2.3 A
Saturation current base	3.93 A
DC resistance base	36 mΩ
Leakage inductance base	0.18 μH
L_f	33 μH
C_f	340 μF

12.5 μs) are shown in figure 4. Figure 5 shows the dynamic of the power during the equalization process, while figure 6 reports the overall efficiency of the proposed balancing circuit considering the multi-winding transformer power loss, P_{tr} , the switching and conduction losses of H-bridges, respectively P_{sw} and P_{on} . The equalization efficiency depends on the power processed by balancing circuit and it is higher than 87.2% with a peak equal to 88.9%. Figure 7 reports the power losses among the components in absolute value and in percentage of the total power losses P_T .

IV. EXPERIMENTAL RESULTS

The validation of the proposed active balancing circuit has been performed on a full-scale prototype. The H-bridges are based on Infineon MOSFET IRLB8721PBF1 with bootstrap driver IR2104STRPBF. For each PWM signal, the isolation between the control unit and each H-bridge has been achieved by an optocoupler (HCPL-3120). The Würth Elektronik 749196547 multi-winding transformer has been selected and operated at 80 kHz. The parameters of components are reported in the Table I. Figure 8 shows a picture of the prototype of MAB active balancing circuit.

To evaluate the performance of the proposed circuit and control, two case scenarios have been considered. In case 1, cell 1 has the higher SoC , whereas cells 2, 3, and 4 have almost the same value of SoC . In the idle operation, the voltage level determines that cell 1 is transferring energy to cells 2, 3, and 4 that are charging. Figures 9 and 10 respectively show the waveforms of the cell voltages and currents during the balancing process. The overall efficiency measures 86 % when $\{V_{B,1}=4.015 \text{ V}, V_{B,2}=3.719 \text{ V}, V_{B,3}=3.689 \text{ V}, V_{B,4}=3.629 \text{ V}\}$ and $\{I_{B,1}=1.058 \text{ A}, I_{B,2}=-0.2414 \text{ A}, I_{B,3}=-0.3065 \text{ A}, I_{B,4}=-0.4482 \text{ A}\}$. The measured efficiency value for this operating point is in line with the results achieved from the numerical

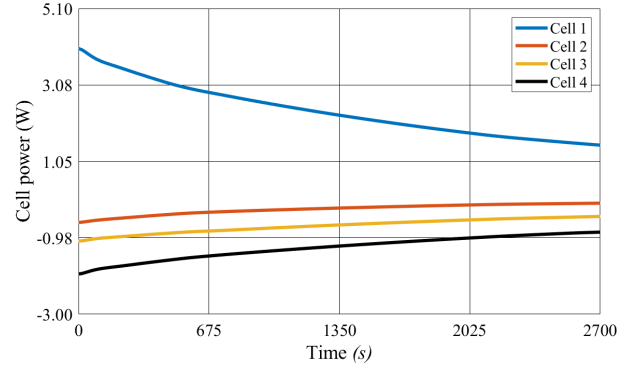


Fig. 5. Numerical results: power of each cell during the equalization process.

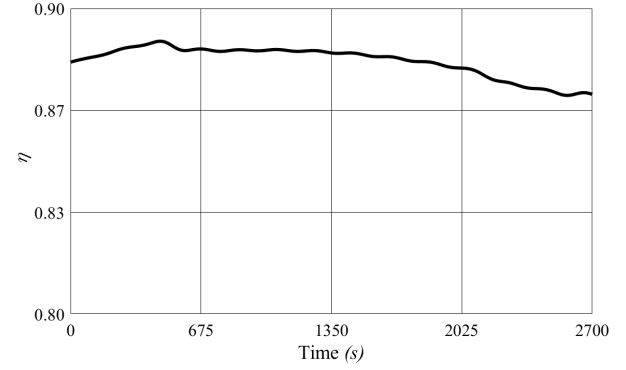


Fig. 6. Numerical results: efficiency during the equalization process.

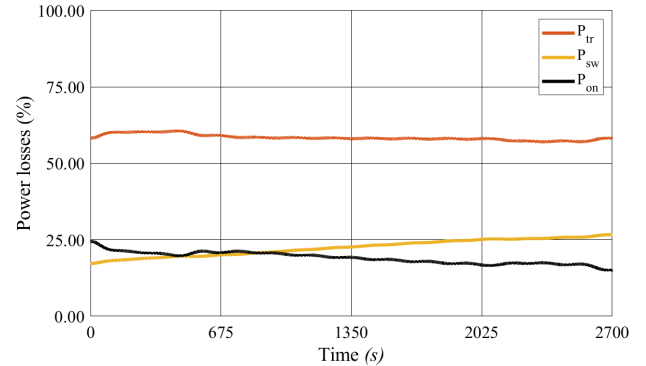
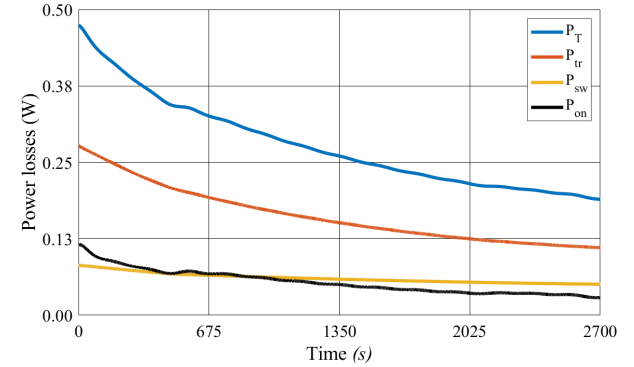


Fig. 7. Numerical results: power losses during the equalization process expressed in absolute value and percentage.

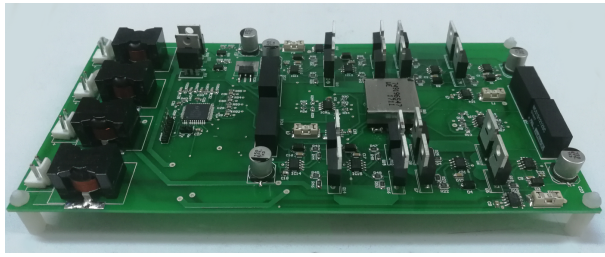


Fig. 8. Picture of the prototype of MAB active balancing circuit.

analysis (section III). The cell voltages and currents for case 2 are depicted in figures 11 and 12, respectively. In case 2, two cells have a higher *SoC* when compared to the others. The experimental results confirm the validity of the proposed active balancing circuit that simultaneously allows charging cells with lower voltage taking energy from cells with higher values of voltage until reaching the equalization balance.

V. CONCLUSION

In this paper, a high-performance equalization circuit based on a Multi Active Bridge for battery active balancing circuit is presented. The proposed structure allows performing a simultaneously energy transfer among the cells of the battery pack. A simple and suitable control algorithm has been developed with the aim of achieving the best performance in all operating conditions. Numerical and experimental results fully confirm the validity of the proposed solution.

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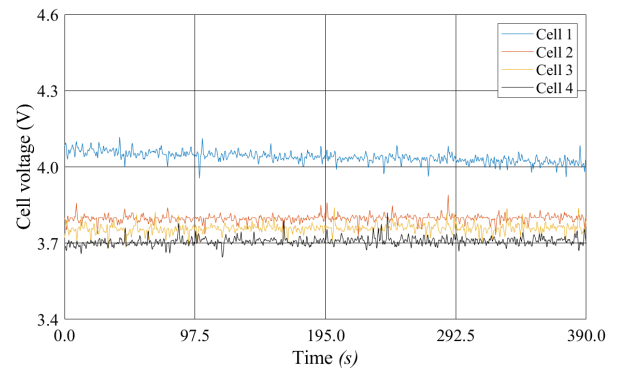


Fig. 9. Experimental results - case 1: cell voltages during the equalization process.

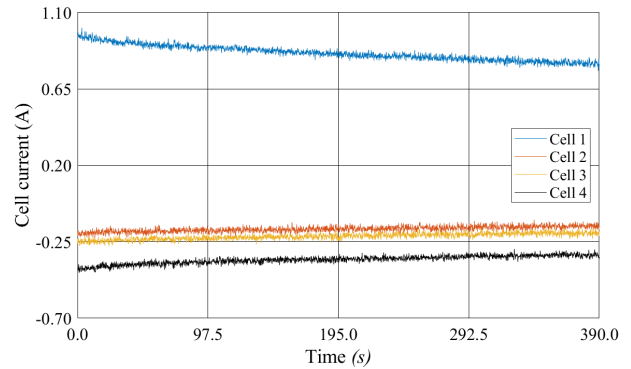


Fig. 10. Experimental results - case 1: cell currents during the equalization process.

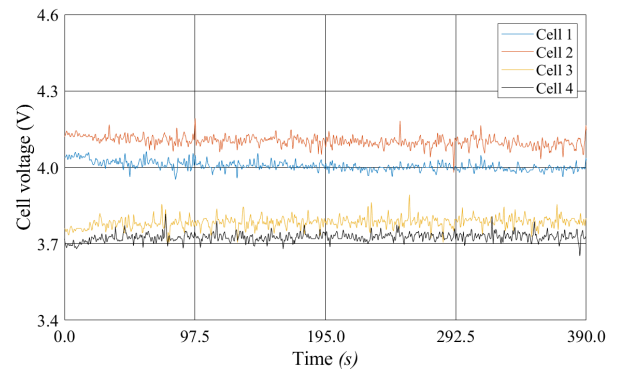


Fig. 11. Experimental results - case 2: cell voltages during the equalization process.

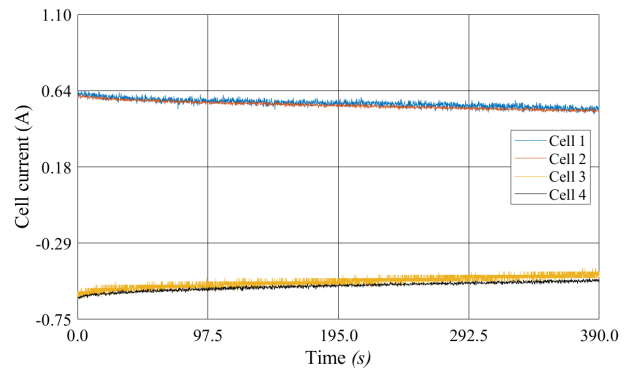


Fig. 12. Experimental results - case 2: cell currents during the equalization process.

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